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main.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#define MAX_FRAMES 50
void sender(int frames[], int size, int window_size);
void receiver(int frames[], int start, int window_size);
int main() {
    int n, window_size;
    int frames[MAX_FRAMES];
    printf("Sliding Window Protocol Simulation (Go-Back-N)\n");
    printf("Enter number of frames to send (max %d): ", MAX_FRAMES);
    scanf("%d", &n);
    if(n > MAX_FRAMES || n <= 0) {
        printf("Invalid number of frames.\n");
        return 0;
    }
    printf("Enter window size: ");
    scanf("%d", &window_size);

    if(window_size <= 0 || window_size > n) {
        printf("Invalid window size.\n");
        return 0;
    }
    // Fill frames with sequence numbers
    for(int i=0; i<n; i++) {
        frames[i] = i;
    }

    sender(frames, n, window_size);

    return 0;
}

void sender(int frames[], int size, int window_size) {
    int base = 0;
    int next_seq = 0;
    char choice;

    srand(time(NULL));

    while(base < size) {
        printf("\nSender window: ");
        for(int i=base; i<base+window_size && i<size; i++) {
            printf("[%d] ", frames[i]);
        }
        printf("\nSending frames...\n");

        for(int i=base; i<base+window_size && i<size; i++) {
            printf("Sent frame: %d\n", frames[i]);
        }

        printf("\nSimulating receiver...\n");
        receiver(frames, base, window_size);

        printf("\nDid all frames receive correctly? (y/n): ");
        scanf(" %c", &choice);

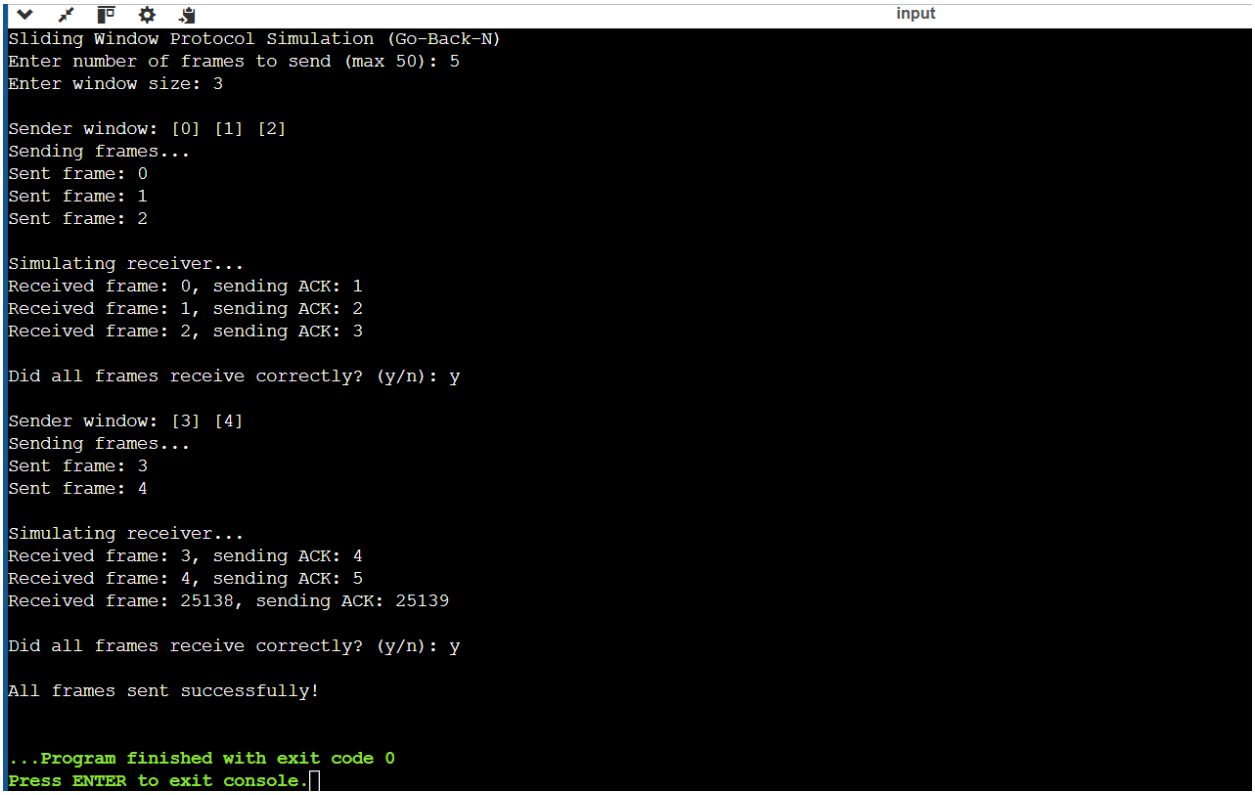
        if(choice == 'y' || choice == 'Y') {
            base += window_size;
        } else {
            printf("Resending from frame %d...\n", frames[base]);
            // In Go-Back-N, we resend from base
        }
    }

    printf("\nAll frames sent successfully!\n");
}

void receiver(int frames[], int start, int window_size) {
    for(int i=start; i<start+window_size; i++) {
        printf("Received frame: %d, sending ACK: %d\n", frames[i], frames[i]+1);
    }
}
```

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main.c

Run

```c
#include <stdio.h>
#include <stdlib.h>
struct packet
{
    int time;
    int size;
} p[50];
int main()
{
    int i, n, m, k = 0;
    int bsize, bfilled, outrate;
    printf("Enter the number of packets:");
    scanf("%d", &n);
    printf("Enter packets in the order of their arrival time\n");
    for (i = 0; i < n; i++)
    {
        printf("Enter the time and size:");
        scanf("%d%d", &p[i].time, &p[i].size);
    }
    printf("Enter the bucket size:");
    scanf("%d", &bsize);
    printf("Enter the output rate:");
    scanf("%d", &outrate);

    m = p[n - 1].time;
    i = 1;
    k = 0;
    bfilled = 0;
    while (i <= m || bfilled != 0)
    {
                k = k + 1;
            }
            else
            {
                printf("\n%dbyte packet is discarded", p[k].size);
                k = k + 1;
            }

        }

    if (bfilled == 0)
    {
        printf("\nNo packets to transmitte");
    }
    else if (bfilled >= outrate)
    {
        bfilled = bfilled - outrate;
        printf("\n%dbytes transfered", outrate);
    }

    else
    {
        printf("\n%dbytes transfered", bfilled);
        bfilled = 0;
    }
    printf("\nPackets in the bucket%dbyte", bfilled);
    i++;
    }
    return 0;
}
```

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Output

```
Enter the number of packets:4
Enter packets in the order of their arrival time
Enter the time and size:1 4
Enter the time and size:2 5
Enter the time and size:3 8
Enter the time and size:4 2
Enter the bucket size:10
Enter the output rate:3


At time 1
4byte packet is inserted
3bytes transfered
Packets in the bucket1byte

At time 2
5byte packet is inserted
3bytes transfered
Packets in the bucket3byte

At time 3
8byte packet is discarded
3bytes transfered
Packets in the bucket0byte

At time 4
2byte packet is inserted
2bytes transfered
Packets in the bucket0byte
```

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```c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

#define BUCKET_SIZE 10
#define PACKET_RATE 4

int bucket = 0;

void addTokens(int tokens)
{
    bucket += tokens;

    if (bucket > BUCKET_SIZE)
    {
        bucket = BUCKET_SIZE;
    }
}

void processPacket(int packet_size)
{
    if (packet_size <= bucket)
    {
        bucket -= packet_size;

        printf("Processed packet of size %d.\n", packet_size);
        printf("Tokens left: %d\n", bucket);
    }
    else
    {
        printf("Not enough tokens to process packet of size %d. "
               "Tokens left: %d\n", packet_size, bucket);
    }
}

```

```c
    }
}

int main()
{
    int incomingPackets[] = {2, 3, 5, 7, 1, 4};

    int n = sizeof(incomingPackets) / sizeof(incomingPackets[0]);

    for (int i = 0; i < n; i++)
    {
        printf("\nTime %d seconds:\n", i + 1);

        addTokens(PACKET_RATE);

        printf("Added %d tokens.\n", PACKET_RATE);
        printf("Bucket now contains %d tokens.\n", bucket);

        processPacket(incomingPackets[i]);

        sleep(1);
    }

    return 0;
}
```

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```
Tokens left: 2

Time 2 seconds:
Added 4 tokens.
Bucket now contains 6 tokens.
Processed packet of size 3.
Tokens left: 3

Time 3 seconds:
Added 4 tokens.
Bucket now contains 7 tokens.
Processed packet of size 5.
Tokens left: 2

Time 4 seconds:
Added 4 tokens.
Bucket now contains 6 tokens.
Not enough tokens to process packet of size 7. Tokens left: 6

Time 5 seconds:
Added 4 tokens.
Bucket now contains 10 tokens.
Processed packet of size 1.
Tokens left: 9

Time 6 seconds:
Added 4 tokens.
Bucket now contains 10 tokens.
Processed packet of size 4.
Tokens left: 6


...Program finished with exit code 0
Press ENTER to exit console.
```